

The Reciprocal-Sum Neumann Inequality: Literature Resolution of an Open Problem in arXiv:1604.05072

Run fa_banach_001 — literature-status note

August 11, 2026

Status. `literature_already_answered`. This note identifies an exact later solution. It does not claim a new theorem.

The source problem

Brasco and De Philippis use the convention

$$0 = \mu_1(\Omega) < \mu_2(\Omega) \leq \mu_3(\Omega) \leq \cdots$$

for the Neumann spectrum. In the subsection “A quick overview of the Neumann spectrum,” they ask whether the higher-dimensional analogue of Szegő’s planar inequality is valid:

$$\frac{1}{|\Omega|^{2/N}} \sum_{k=2}^{N+1} \frac{1}{\mu_k(\Omega)} \geq \frac{1}{|B|^{2/N}} \sum_{k=2}^{N+1} \frac{1}{\mu_k(B)}. \quad (\text{S})$$

Here $B \subset \mathbb{R}^N$ is a ball. Because its first nonzero eigenvalue has multiplicity N , the right side is $N/(|B|^{2/N} \mu_2(B))$. The survey says explicitly that the validity of (S) was open [1].

Exact later theorem

He, Li, and Tang number the Neumann eigenvalues instead as

$$0 = \mu_0(\Omega) < \mu_1(\Omega) \leq \mu_2(\Omega) \leq \cdots.$$

Their main result is the following [2, Theorem 1.2].

Theorem. *Let $N \geq 2$, let $\Omega \subset \mathbb{R}^N$ be a bounded domain with smooth boundary, and let B be a ball with $|B| = |\Omega|$. Then*

$$\sum_{j=1}^N \frac{1}{\mu_j(\Omega)} \geq \frac{N}{\mu_1(B)}.$$

Equality holds if and only if Ω is a ball.

This is the Ashbaugh–Benguria conjecture. Li and Wang subsequently record that the proof remains valid for every bounded Lipschitz domain and state the result in that form [3, Theorem 1.1]. They also prove stronger quantitative stability estimates for the same deficit.

Translation to the source notation

The index shift is the only spectral difference: the later μ_j is the source's μ_{j+1} . To check the normalization, let $\mathbb{B}$ be the unit ball, let

$$R_\Omega = \left(\frac{|\Omega|}{|\mathbb{B}|} \right)^{1/N}, \quad \tilde{\mu}_j(\Omega) = R_\Omega^2 \mu_j(\Omega).$$

Li and Wang state the theorem as

$$\sum_{j=1}^N \frac{1}{\tilde{\mu}_j(\Omega)} \geq \frac{N}{\mu_1(\mathbb{B})}. \quad (1)$$

Since $R_\Omega^{-2} = (|\mathbb{B}|/|\Omega|)^{2/N}$, division of (1) by $|\mathbb{B}|^{2/N}$ gives

$$\frac{1}{|\Omega|^{2/N}} \sum_{j=1}^N \frac{1}{\mu_j(\Omega)} \geq \frac{N}{|\mathbb{B}|^{2/N} \mu_1(\mathbb{B})}.$$

Scale invariance and the N -fold equality $\mu_1(B) = \dots = \mu_N(B)$ turn the right side into

$$\frac{1}{|B|^{2/N}} \sum_{j=1}^N \frac{1}{\mu_j(B)}.$$

After shifting j to $k = j + 1$, this is exactly (S). The equality statement also transfers: equality occurs precisely for balls.

For a disconnected Lipschitz open set, the source convention supplies additional zero eigenvalues; the reciprocal expression is then infinite (or is excluded from the finite formulation). Thus the substantive connected case is exactly the domain case covered above.

Proof provenance and scope

The later proof is not a citation-level restatement. It couples the N trial functions transplanted from the first eigenspace of the ball, applies a trace-form Ritz/Hersch variational principle, and proves the needed matrix comparison by exploiting a trace-free angular-imbalance matrix. Equality in the matrix and raywise rearrangement steps forces the domain to be a ball [2].

This packet resolves only the displayed reciprocal-sum Neumann problem. The same source also asks for the sharp topology-free upper bound on $P(\Omega)\sigma_2(\Omega)$ for planar Steklov eigenvalues. That is a separate question and is not claimed to follow from the Neumann theorem.

References

- [1] L. Brasco and G. De Philippis, *Spectral inequalities in quantitative form*, arXiv:1604.05072 (2016), Section 4.
- [2] Y. He, Y. Li, and Q. Tang, *A proof of the Ashbaugh–Benguria conjecture for reciprocal sums of Neumann eigenvalues*, arXiv:2606.08271 (2026).
- [3] Y. Li and Z. Wang, *Spectral and Geometric Stability for the Reciprocal Sum of Neumann Eigenvalues*, arXiv:2607.19008 (2026).